TDD in the Agentic Era

What we keep, what we leave behind

Sebastian Larsson

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WHO AM I

Sebastian Larsson

/ Developer

/ Trainer

/ Manager

12 years teaching TDD

> **“I’m not a great programmer; I’m just a good programmer with great habits.”**
>
> KENT BECK

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TDD is a habit. Agents don’t have habits, they have instructions.

I no longer think agents should do TDD the way I taught it.

PART 1

The old days

What TDD is, and why it works for humans

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THE OLD DAYS

TDD myths

~~"TDD is writing the tests first"~~

~~"TDD is about testing"~~



"You don't write 'the tests' first. You write a test first."

TIM OTTINGER

"TDD is not really about writing tests before code. It's about using tests in short-lived RGR cycles as a tool to design code."

ANDREA LAFORGIA

"The tests that drive our design often correspond to good regression tests, but certainly not always."

JASON GORMAN

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THE OLD DAYS

Definition

→ A development technique

→ Breaking a problem into pieces

→ Rapid feedback

→ A design process

Design results

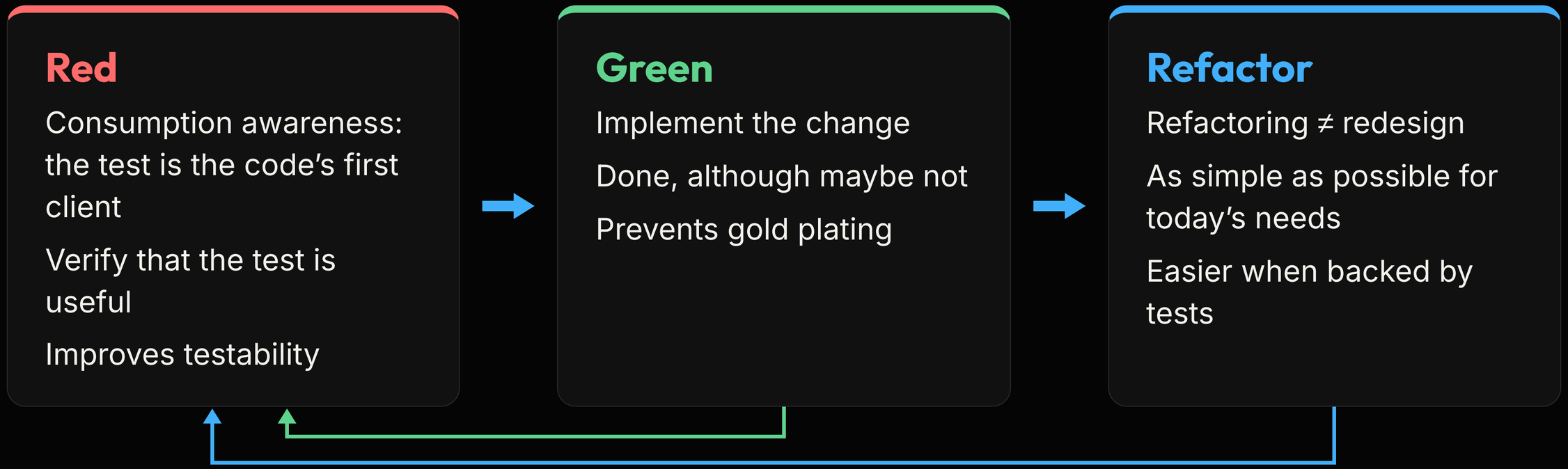
Dependencies are injected

A modular and extensible system

Frequent use of composed method

THE OLD DAYS

Why TDD works for humans



Each step answers a human limitation.

THE OLD DAYS

Red: create a failing test



Consumption awareness

Verify that the test is useful

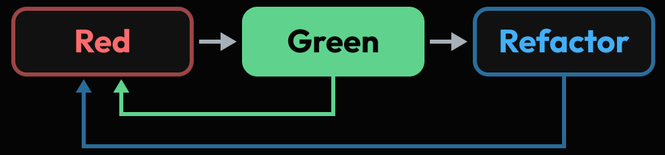
Improves testability

```
@Test
public void simpleAddition() {
    Calculator calculator = new Calculator();
    int expected = 3;
    int actual = calculator.add(1, 2);
    assertThat(actual, is(expected));
}
```

```
@Test
public void detects_adjacent_mine() {
    Mineplacer mineplacer = new Mineplacer() {
        public String constructMineField() {
            return "[*,*,*,*]"
               + "[*,X,*,*]"
               + "[*,*,*,*]"
               + "[*,*,*,*]";
        }};
    MineField mineField = new MineField(mineplacer);
    assertEquals(1, mineField.atPos(0, 0));
}
```

THE OLD DAYS

Green: make the test pass



- → Implement the change
- → Means done, although maybe not
- → Prevents gold plating



THE OLD DAYS

Gold plating

```java
interface MessageStrategy {
    public void sendMessage();
}
abstract class AbstractStrategyFactory {
    public abstract MessageStrategy createStrategy(MessageBody mb);
}
class MessageBody {
    Object payload;
    public Object getPayload() {
        return payload;
    }
    public void configure(Object obj) {
        payload = obj;
    }
    public void send(MessageStrategy ms) {
        ms.sendMessage();
    }
}
public class MyProgram {
    public static void main(String[] args) {
        MessageBody mb = new MessageBody();
        mb.configure("Hello World!");
        AbstractStrategyFactory asf = DefaultFactory.getInstance();
        MessageStrategy strategy = asf.createStrategy(mb);
        mb.send(strategy);
    }
}
class DefaultFactory extends AbstractStrategyFactory {
    private DefaultFactory() { }
    static DefaultFactory instance;

    public static AbstractStrategyFactory getInstance() {
        if (instance == null)
            instance = new DefaultFactory();
        return instance;
    }
    public MessageStrategy createStrategy(final MessageBody mb) {
        return new MessageStrategy() {
            MessageBody body = mb;
            public void sendMessage() {
                Object obj = body.getPayload();
                System.out.println((String) obj);
            }
        };
    }
}
```

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lines of Java to print “Hello World!”, with a strategy, an abstract factory and a singleton

THE OLD DAYS

Refactor as needed

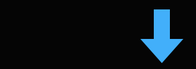


→ Refactoring ≠ redesign

→ As simple an implementation as possible for today's needs

→ Easier when backed by tests

```
// ...
Person p = getPerson(socialSecNr);
//print the person
System.out.println(p.getFirstName());
System.out.println(p.getLastName());
System.out.println(p.getSocialSecNr());
```



```
// ...
Person person = getPerson(socialSecNr);
print(person);
```

BRAIN DUMPING GROUND

Get it out of your head and onto the list.

Limited working memory

→ Executable examples

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MERCILESS REFACTORING

~~FEAR~~

The tests have your back.

- → Freedom to refactor
- → Safely make changes

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THE OLD DAYS

The todo list

→ A dynamic buffer that keeps noise away

→ Examples of test cases

→ **Test order has a huge impact on TDD flow**

TODO

- [ ] Equal cards: Tie
- [ ] Steve wins 1 to 0
- [ ] Josh wins x to y
- [ ] J beats T
- [ ] A beats K

THE OLD DAYS

The evidence for humans

40–90%

fewer pre-release defects in four industrial teams (Nagappan et al. 2008, Microsoft and IBM)

15–35%

more development time, as the teams themselves estimated it

→ Fucci et al. 2017: short, steady cycles mattered more than writing the test first

→ Ghafari et al. 2020: studies disagree mostly because they’re set up differently

Every step answers a human limitation. Which of them does an agent actually have?

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LIVE DEMO

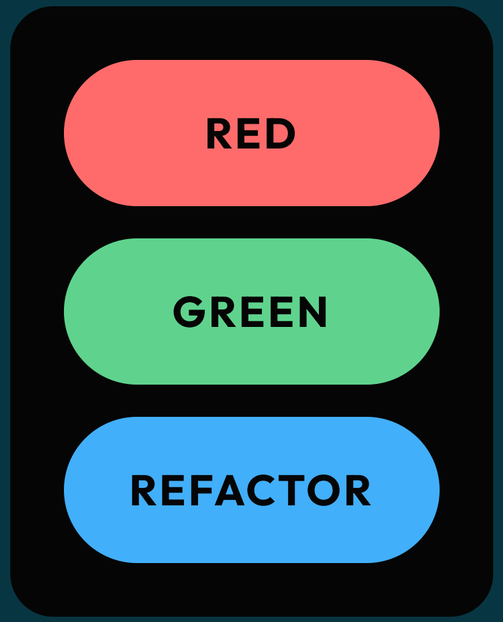
The cards game

codewars.com/kata/53417de006654f4171000587

Steve vs Josh: the higher card wins the round.

“Steve wins x to y”, “Josh wins x to y” or “Tie”

Same kata, two agents: one told to do TDD, one not. Let’s race.



PART 2

The new ways

What happened when agents were told to do TDD

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THE NEW WAYS · AN AGENT ≠ A HUMAN

Which of TDD's benefits does an agent need?

It doesn't have

Fear

Boredom

Not knowing where the code is going

It has, as badly as us or worse

Misreading the spec

Claiming done when it isn't

Blind spots in its own tests

Gold plating: Beck warns about unrequested features

GLM wrote the same misreading into its test and its code.

THE NEW WAYS

The experiment

9 + 1

agent runs on the same kata: 9 told to do TDD, 1 told not to

2016

my own run, live in front of an audience

9

planted bugs via mutation testing, plus an acceptance suite the agents never saw

→ Every red and green replayed from the git history

THE NEW WAYS

Agentic TDD skills: it looks good, but it’s a charade

46%

of tests passed as soon as they were written (26 of 57): the agent already knew the answer

→ GLM: 7 of 7 genuine reds, every commit labelled correctly, and it ships “Josh wins 1 to 2”

→ Ten implementations, one design: a rank string, indexOf, two counters

→ By Laforgia’s definition (tests as a tool to design code), the agents did test-first, not TDD

→ Reflections: one copied from another model, one cites commit hashes that don’t exist

Did the design emerge?

Design result	What the agents did
Composed method	Yes, in the TDD runs' refactor steps. The no-TDD run is one function.
Dependencies injected	No. Only my 2016 run injected the ranking.

OPUS, AFTER ITS REFACTOR STEPS

```
return formatResult(scoreRounds(steve, josh));
```

THE NEW WAYS

Tiny steps aren’t meaningful for an agent

TDD as if you meant it	Tiny steps, the code grows inside the test (my 2016 run)
Fake it	Adds bugs on purpose: “TDD created a bug and then caught it” (GLM)
Minimal steps	The weakest frontier-model suite: Opus 5 caught 5 of 9 planted bugs
Gorman called it	The tests that drive the design aren’t always good regression tests
“A test first, not the tests”	A human rule: GLM wrote all 7 tests up front as skipped tests, and for an agent that was fine

THE NEW WAYS

The cost: tokens, dollars and time

Run	Time	Cost	Planted bugs caught
No TDD · Opus 5.5	29 s	$0.27	7 of 9
TDD by prompt · Opus 5, run 1	3 min 47 s	$2.07	8 of 9
TDD by prompt · Opus 5, run 2	4 min 18 s	$2.72	5 of 9
TDD enforced by probity · Opus 5.5	10 min 48 s	$3.54	6 of 9

8–13× the cost and 8–22× the time, with no measurable quality gain.

THE NEW WAYS

What others found

BIRGITTA BÖCKELER · AUG 2026

TDD prompts gave no quality gain, for 3–8.5× the tokens.

EMILY BACHE · SEP 2026

That tested a prompt, not TDD. A test list and a harness are what's missing.

→ **So I ran it again with a harness.**

THE NEW WAYS

Enforcing TDD with a harness: probity

11 of 12

tests genuinely red

4

real refactors: the most faithful TDD in the repo

→ The same result as no TDD, at 13× the cost and 22× the time

→ It dropped the tests that couldn’t go red, so it caught fewer planted bugs: 6 of 9, against 7 of 9 without TDD

→ 9 of 13 blocks were false positives; resending the same edit usually got through

Are harnesses worth the cost?

Model	Harness	Accepted	Time	Cost
Sol (frontier)	No	10 of 10	6 min 51 s	~$0.41
Sol (frontier)	Yes	10 of 10	19 min 53 s	~$2.68
Luna (cheap)	No	3 of 10	15 min 56 s	~$0.09
Luna (cheap)	Yes	10 of 10	35 min	~$0.27

→ His harness isn’t TDD: human-written specs and examples, plus an independent acceptance gate

The ceremony buys nothing for a frontier model on a task it can already do.

LIVE DEMO

The herdr race: results

TDD agent

TIME	[__ min]
COST	[$__]
ACCEPTANCE SUITE	[_ of 8]

No-TDD agent

TIME	[__ min]
COST	[$__]
ACCEPTANCE SUITE	[_ of 8]

PART 3

The way forward

For the coming year

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THE WAY FORWARD

Leave behind

~~Fake it and minimal green~~

~~Forcing the smallest step~~

~~A perfect process as proof of quality~~

~~Red as the proof that a test can fail~~

→ **Keep the proof, drop the ritual: mutation-test afterwards instead (Gorman, Böckeler)**

THE WAY FORWARD

Move up a loop



The double loop from Growing Object-Oriented Software (Freeman & Pryce, 2009)

→ Outer loop, acceptance tests: the human

→ Inner loop, red/green/refactor: the agent

Isn’t this just ATDD/BDD again? Yes, and that’s the point.

THE WAY FORWARD

Planning

Why? To not have to redo, and to make sure interpretations are correct.

The ambiguity

GLM's "Josh wins 1 to 2": no agent's TDD caught the "x to y" ambiguity.

A human reading the test list would have caught it in seconds.

The order

The only reds that changed the code came from face cards after number cards (Opus 5: A beats K, Astra: J beats T).

The human picks and orders the examples.

A small step compared to spec-driven development (spec-kit, OpenSpec)

THE WAY FORWARD

Use acceptance-test-driven examples to specify desired behaviour

THIS EXPERIMENT

Only the acceptance suite the agents never saw caught GLM’s and Gemma’s bugs

MATHEWS & NAGAPPAN 2024

Tests given with the task help LLMs

KONSTANTINOU ET AL. 2026

14% vs 25%

faults found by tests written after faulty code vs. independent tests

→ Jain’s harness is the same idea, tested at scale

Examples are executable; a spec document isn’t.

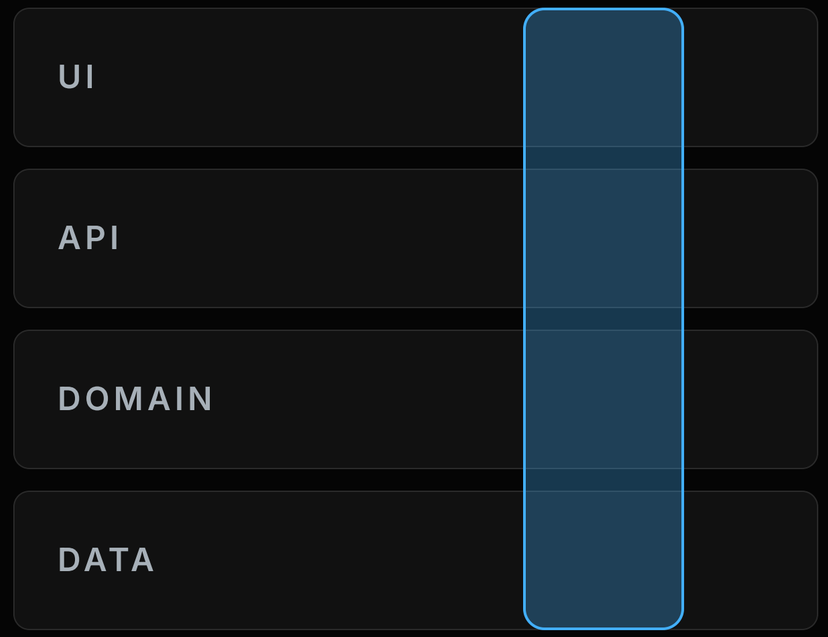
THE WAY FORWARD

Vertical slicing

→ Small diffs a human can actually review

→ Fast incremental delivery at all layers

From experience, not from this experiment



ONE SLICE THROUGH EVERY LAYER

Test-first becomes spec-first.

From Beck’s TDD by Example to Adzic’s Specification by Example.

The human owns the examples. The agent does the cycles.

Questions?

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